Math 526, S05 **Name:**

Quiz 10 **SSN: xxx-xx-**

<u>Instructions:</u> There are totally 1 problem. You must show all of your work to receive a credit for a correct answer. No graphical calculator is allowed in the quiz and exam.

Problem 1. (20 points) Let $\mathbf{A} = \begin{bmatrix} 1 & -1 & 2 \\ 0 & 1 & -2 \\ 0 & -2 & 1 \end{bmatrix}$,

a) Find the eigenvalues of $\mathbf{A}$.

Solution:

$$\begin{aligned} det(\mathbf{A} - \lambda\mathbf{I}) &= 0 \\ det\begin{bmatrix} 1-\lambda & -1 & 2 \\ 0 & 1-\lambda & -2 \\ 0 & -2 & 1-\lambda \end{bmatrix} &= 0 \\ (1-\lambda)[(1-\lambda)^2 - 4] &= 0 \\ (1-\lambda)(\lambda^2 - 2\lambda - 3) &= 0 \\ (1-\lambda)(\lambda+1)(\lambda-3) &= 0 \\ \lambda &= 1, -1, 3 \end{aligned}$$

b) Compute $tr\mathbf{A}$. What is relation between $tr\mathbf{A}$ and the eigenvalues?

Solution: $tr\mathbf{A}$ is the sum of the diagonal components of $\mathbf{A}$, so

$$tr\mathbf{A} = 1 + 1 + 1 = 3$$

In fact, $tr\mathbf{A}$ also equals the sum of all eigenvalues $\mathbf{A}$, i.e,

$$tr\mathbf{A} = 1 + (-1) + 3 = 3$$